



Research papers

Circulations associated with cold pools in the Bohai Sea on the Chinese continental shelf



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ABSTRACT

Circulations associated with cold pools in the Bohai Sea were investigated in the summers of 1959 and 2006 using observations and numerical simulations. The climatological forcing was incorporated to drive a model based on the Regional Ocean Modeling Systems (ROMS) to reconstruct the monthly evolution of flow structure. Cold pools were found to be separated into two parts by an undersea ridge in the central Bohai Sea. A set of numerical experiments were conducted to evaluate the relative contributions from the exchange across the strait, heat flux, wind, and tides. Baroclinic circulation was the dominant component of the circulation. A basin-scale cyclonic gyre was the main feature during the stratified seasons from May through August. Cyclonic and anticyclonic sub-basin scale gyres emerged in late spring and peaked during July and early August, accompanied by intensifying frontal jets along the west and south coasts of the Bohai Sea. The cyclonic frontal jet in the model was responsible for rapid reduction of particles released from the surface layer of the Bohai Sea in summer; and 2) Experiments using Lagrangian particle tracking also showed that particles released in Liaodong and Bohai bays were retained longer than those released in the other regions as the surface convergence in the western Bohai Sea resulted in an intense concentration of particles.

1. Introduction

Cold pools are isolated low-temperature areas and often form dense water masses below the seasonal thermocline. They are usually associated with topographic depressions. For example, cold pools have been observed worldwide including the Yellow Sea (He et al., 1959; Su and Huang, 1995; Pang et al., 2004; Wang et al., 2014), the western Irish Sea (Hill et al., 1994, 2008), the North Sea (Brown et al., 1999), Bay of Fundy (Aretxabaleta et al., 2008), and the Bering Sea shelf (Reed and Staben, 1996; Zhuang et al., 2014). Strong horizontal temperature gradients separate the cold pools from the well-mixed regions of water bodies (Simpson et al., 1978). Using an analytical model, Hill (1996) suggested that cyclonic circulation always occurs near the surface around dome-like cold pool. A circular frontal jet has also been observed from field surveys (Hill et al., 1997; Brown et al., 1999; Horsburgh et al., 2000; Lin et al., 2002; Pang et al., 2004).

The presence of cold pools in the Bohai Sea has been well recognized. Three shallow bays, Liaodong, Bohai and Laizhou, form the Bohai Sea, which is a semi-enclosed shelf sea connected to the Yellow Sea through the Bohai Strait (Fig. 1). A subsurface Central Bank dissects the three bays in the central Bohai Sea. The bowl-shaped northeastern depression is enclosed by the 30-m isobath, while the band-shaped southwestern depression lies in the southern Bohai Sea and is slightly deeper than 25 m. During stratified periods of the year, typically from late April to mid-September (Bao et al., 2004; Wu et al., 2004), cold pools were frequently observed in the Bohai Sea (Liu et al., 2003; Wan et al., 2004; Lin et al., 2006; Jia et al., 2008). Bottom temperature in the northern region appears to be 1–3 °C lower than that in the south because of persistent cold water advection from the Yellow Sea along the east coast of the Bohai Sea (Zhou et al., 2009). The Laotieshan Channel forms the main entrance through which cold water enters from the northern Yellow Sea (Feng, 1987).

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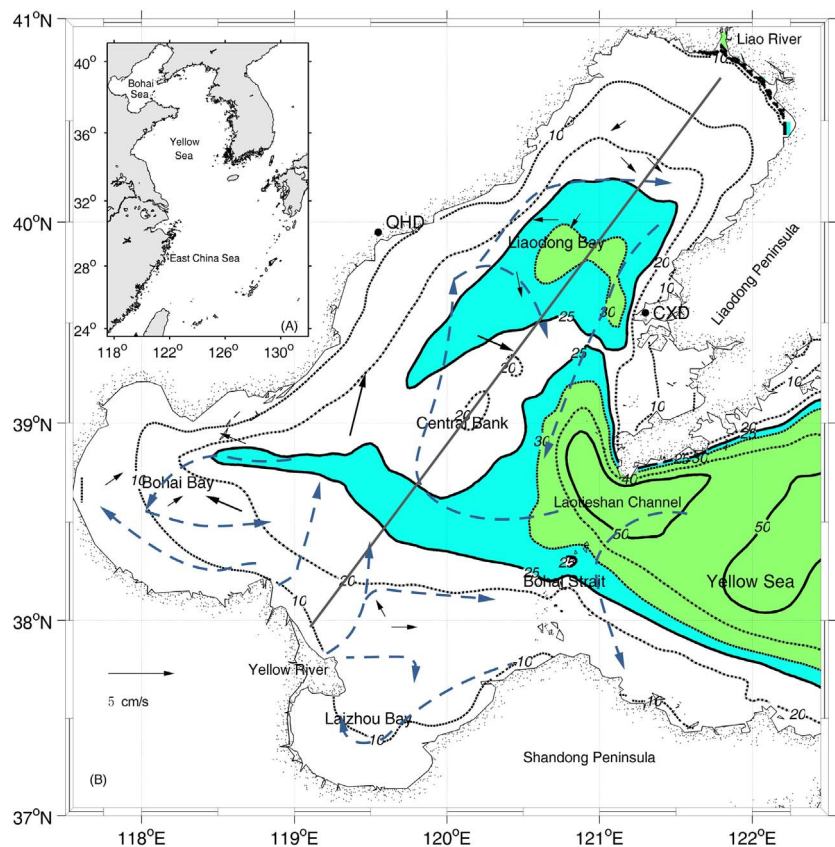


Fig. 1. (a) Model domain and (b) topographic features of the Bohai Sea. Isobaths of 10, 20, 25, 30, and 50 m are indicated by contours, while topographic depressions are emphasized by shadings. Solid arrows are depth-averaged flow measurements from oil platforms, and dashed arrows are schematic circulations (after Zhao et al., 1995). Traditional Liaodong Bay represents the region in the north of Qinhuangdao (QHD) and Changxindao (CXD). In the paper, Liaodong Bay refers to the region north of the Central Bank. The transect (solid line) was highlighted as an indicator of the cold/dense pools in previous studies, e.g., Liu et al. (2003) and Lin et al. (2006).

Although the spatio-temporal characteristics of the cold pools and overlying stratification in the Bohai Sea have been intensively investigated (Huang et al., 1999; Zhao et al., 2001; Bao et al., 2004; Liu, 2007), knowledge of the corresponding circulations and their contributions to the basin-scale gyre remains unclear. Based on hydrographic observations, Guan (1994) suggested there were a basin-scale counter-clockwise circulation and two cyclonic cells above the two depressions. In contrast, Zhao et al. (1995) documented a clockwise-dominant flow pattern in the northern Bohai Sea based on current observations taken from oil platforms (see dashed arrows in Fig. 1b). Most historical hydrographic and current data available were the results of episodic monitoring programs or occasional research cruises; therefore, they were limited in temporal or spatial extent and, consequently, in the amount of information they could provide.

The bay experiences weak tidal currents except in the Bohai Strait (Ogura, 1936) and along the coast of Liaodong Bay (Qiao et al., 2006). The average amplitude of M_2 barotropic tide in the three bays is listed in Table 1. Residual tidal currents over most the basin are essentially less than 2 cm/s according to model simulations (Fang and Yang, 1985; Huang et al., 1999). Monsoon wind plays an overwhelming role on the circulation in winter (Miao and Liu, 1988), while the rather weak winds in summer plays a less important role in driving the circulation except at the surface (Zhao and Shi, 1993). Recent modeling studies suggested

the importance of a horizontal density gradient to the basin-scale gyre (Liang et al., 2006; Wang et al., 2010). Using a diagnostic model, Wan et al. (2004) recognized the significance of density-driven circulation using a set of experiments based on a hydrographic survey conducted in August 2000. However, the intra-seasonal variability of circulations associated with the cold pools remains unknown.

As a semi-enclosed sea neighboring large human population and rapidly urbanized cities, the Bohai Sea is vulnerable to increasing inputs of nutrients and pollutants (Wei et al., 2002). Zhai et al. (2012) observed a band-like low oxygen zone west of Central Bank, where oxygen depletion rate was larger than the rest of the surveyed region. Only a few studies have addressed biological influences of the cold pools in the Bohai Sea (e.g. Lin et al., 2006), even though similar issues have been extensively addressed in other regions. Pingree et al. (1975) reported high phytoplankton biomass along the shelf sea front in the English Channel. It was suggested that the cyclonic sub-surface gyre in the western Irish Sea was a retaining system for the larvae of Norway lobster (*Nephrops norvegicus*) in spring (Hill et al., 1994, 1997). High concentration of fish larvae around the cold pool in the southern Yellow Sea was also documented (Wei et al., 2003).

This study aims at understanding the contributions of different forcing factors to regional circulations, namely, how will the baroclinicity associated with these cold pools affect the circulations in the

Table 1
Amplitude of M_2 barotropic current in different parts of the Bohai Sea (units: m/s).

Region	Liaodong Bay	Bohai Bay	Laizhou Bay	Central Bank	Bohai Strait	Bohai Sea
Mean	0.29	0.31	0.17	0.28	0.26	0.08
RMS	0.31	0.33	0.20	0.29	0.27	0.09

Bohai Sea? In addition, how will the transport of suspended and dissolved matters be affected by the frontal jet surrounding each cold pool? The period of interest is May to August when stratification is significant.

2. Data and methods

2.1. Hydrographic data

Hydrographic data were collected based on large-scale surveys carried out during July 5–12, 1959 and July 19–August 12, 2006. The two data sets represent the fundamental features of the Bohai Sea in different time periods, separately by the rapid change of the Yellow River freshwater discharge around the 2000s (Mao et al., 2008). All data were collected and processed by the North China Sea Branch of the State Oceanic Administration. For the earlier cruise, temperature data were measured by reversing thermometer, and salinity was converted from chlorinity titration. During the cruise in 2006, both temperature and salinity were measured by a Seabird CTD.

2.2. Circulation model and forcing

The free-surface, hydrostatic, three-dimensional primitive-equation model used here is based on the Regional Ocean Modeling systems (ROMS) (Shchepetkin and McWilliams, 2003, 2005; Wilkin et al., 2005; Haidvogel et al., 2008; Warner et al., 2008). ROMS is solved using a split-mode method to separate the barotropic and baroclinic processes to save model integration time, with a filter to reduce the mismatch between the two modes (Shchepetkin and McWilliams, 2005). The time step for the baroclinic equations is 92 s, which is 30 times larger than that for the barotropic equations. Vertical turbulence closure is the k - ϵ scheme (Rodi, 1987), which is re-formulated by Warner et al. (2005) under the generic length-scale framework of Umlauf and Burchard (2003). The model uses a simple drag-coefficient method with quadratic formulation to calculate the bottom stress with a uniform drag coefficient of 0.003.

The model domain covers a rectangular area (Fig. 1a), with a homogeneous $1/24^\circ$ horizontal resolution in each horizontal dimension. The domain has three open boundaries: the southern boundary at 23.5°N , the eastern boundary at 132°E and the northern boundary at 41°N (north of the Korea Peninsula). There are 30 terrain-following vertical layers in the s -coordinate system similar to that used in Song and Haidvogel (1994). The model was initialized using January data of multi-year observations (Chen, 1992). For simplicity, climatological forcing from the monthly $0.5^\circ \times 0.5^\circ$ Comprehensive Ocean-Atmosphere Data Set (COADS; Da Silva et al., 1994) was periodically used to drive the model for 10 years. The COADS includes net heat flux (Q), shortwave radiation, surface freshwater flux, and wind stress. This data set has the advantage of having a relatively long reference period, which is important for the climatological mean, and has also been validated by modeling studies for the eastern Chinese seas (Huang et al., 2005; Zhou et al., 2009, 2015). Thermal feedback was introduced to correct heat flux with bulk formulas (Marchesiello et al., 2001). Sea surface salinity from Chen (1992) was applied to the model to correct the sea surface freshwater flux using a scheme similar to that suggested by Barnier et al. (1995). The correction of freshwater flux was calculated using $S_m(EP)/\rho_0 + \alpha(S_m - S_{clm})$, where S_m and S_{clm} are simulations and climatological observations of sea surface salinity, respectively; and EP is surface freshwater flux. The nudging coefficient α was chosen for our configuration, corresponding to 30 days. Free surface, velocity, temperature, and salinity at the open boundaries were all extracted from the HYCOM interannual North and Equatorial Pacific Ocean $1/12^\circ$ simulation (Kelly et al., 2007), by averaging outputs from 1979 to 2003, plus M_2 tide from an inverse tide model (Egbert and Erofeeva, 2002). An Orlanski-type radiation condition was applied at the open boundaries for temperature, salinity and baroclinic velocity

(Marchesiello et al., 2001).

The Yellow and Liao rivers are included in the study, in addition to five other rivers outside the Bohai Sea (Zhou et al., 2015). Their temperature and salinity were estimated based on a climatological atlas (Chen, 1992). The river discharges estimated here were similar to those used by Chang and Isobe (2003), except for the Yellow River. This was once the second largest river emptying into the eastern Chinese seas, but its annual-mean flux in the lower reach (Lijin Hydrologic Station) had been reduced dramatically from $1600 \text{ m}^3/\text{s}$ in the 1960s to $420 \text{ m}^3/\text{s}$ in the late 1990s because of increasing human demands for water within the watershed. The discharge of the Liao River is approximately one-fourth that of the Yellow River. Both rivers are considered using their monthly water transports.

2.3. Model experiments

A reference run was conducted for analyzing of the evolution of seasonal stratification and associated circulations. All forcing factors are included in the reference run, e.g., wind (W), tide (T), river runoff (R), and air-sea heat flux (D) (Case T+W+R+D). A set of sensitivity experiments was then run to identify the contributions from individual forcing to the basin-scale gyre. The sensitivity experiments include: Case W+D+R, a simulation with wind, rivers and heat flux, but without tide; Case T+D, a simulation driven by heat flux and tide only; Case T+W+D, a simulation with heat flux, wind and tide only; Case D is only driven by heat flux. One more experiment was carried out to clarify the contribution from the exchange between the Bohai Sea and the Yellow Sea through the Bohai Strait (Case OBC).

While the circulation pattern driven by an individual forcing is unrealistic and hard to verify, the contribution from each forcing to the basin-wide gyre is approximated by calculating the difference between two experimental results following the method presented by Aretxabaleta et al. (2008). For instance, the effects of tidal mixing on the density structure and circulation are estimated by subtracting Case W+D+R from the reference run. The effects of wind are similar to the difference between Case T+D and Case T+W+D. The river runoff effects are close to the subtraction of Case T+W+D from the reference run. For cases without rivers, only river runoff in the Bohai Sea is removed, while the other rivers flowing into the Yellow Sea and East Chinese Sea are still included to match the actual situation. A similar approach is applied to all experiments without wind. Case OBC omits wind, heat flux, rivers around the Bohai Sea, and M_2 over the model domain.

2.4. Lagrangian particle-tracking experiments

An online Lagrangian particle tracking (LPT) method with vertical diffusion was applied to trace water particle trajectories, which represent movements of suspended and dissolved matters. Ocean water has high variability in turbulence and thus leads to a non-uniform vertical diffusion profile especially where stratification is present (Visser, 1997). Therefore, LPT with a diffusive random walk scheme was selected (Hunter et al., 1993). A total of 1129 released neutral particles inside the Bohai Sea were equally distributed at three depths, the surface, 10 m and near the sea bed, which all experienced different baroclinic effects. The neutral condition means all particles were released with zero initial speed.

3. Results

3.1. Observations of temperature and salinity

The relative contributions of temperature and salinity to the seasonal evolution of density field can be clarified based on field observations. As mentioned previously, the temperature pattern was less variable than that of salinity during different years; salinity increased tremendously in the past few decades (Lin et al., 2001;

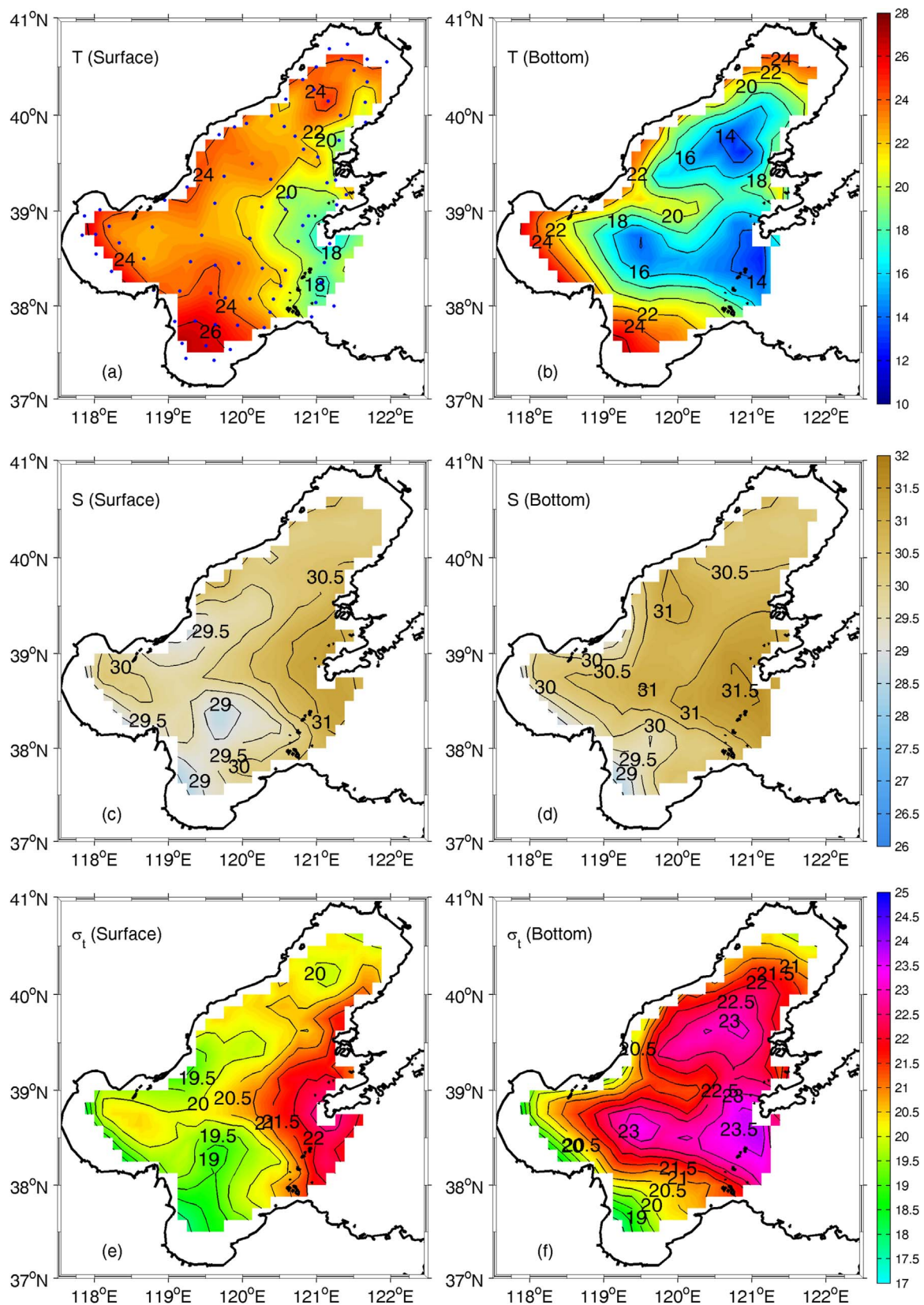


Fig. 2. Distribution of temperature, salinity and density (σ_t) at the surface (left panels) and at the bottom (right panels) based on the cruise in July 1959. The contour interval is 2.0 for temperature, and 0.5 for both salinity and σ_t . Blue dots in the top left panel represent sampling stations. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article).

Fang et al., 2002; Bao et al., 2004; Wu et al., 2004). Figs. 2 and 3 show the spatial distributions of temperature and salinity, respectively, based on the 1959 and 2006 survey results. The similarity between the

temperature and density fields is apparent (Fig. 2b vs 2f, Fig. 3b vs 3f). Particularly, cold water mass ($T < 16^\circ\text{C}$) existed in Liaodong Bay, which is discernable with enclosed contours and shows consistent

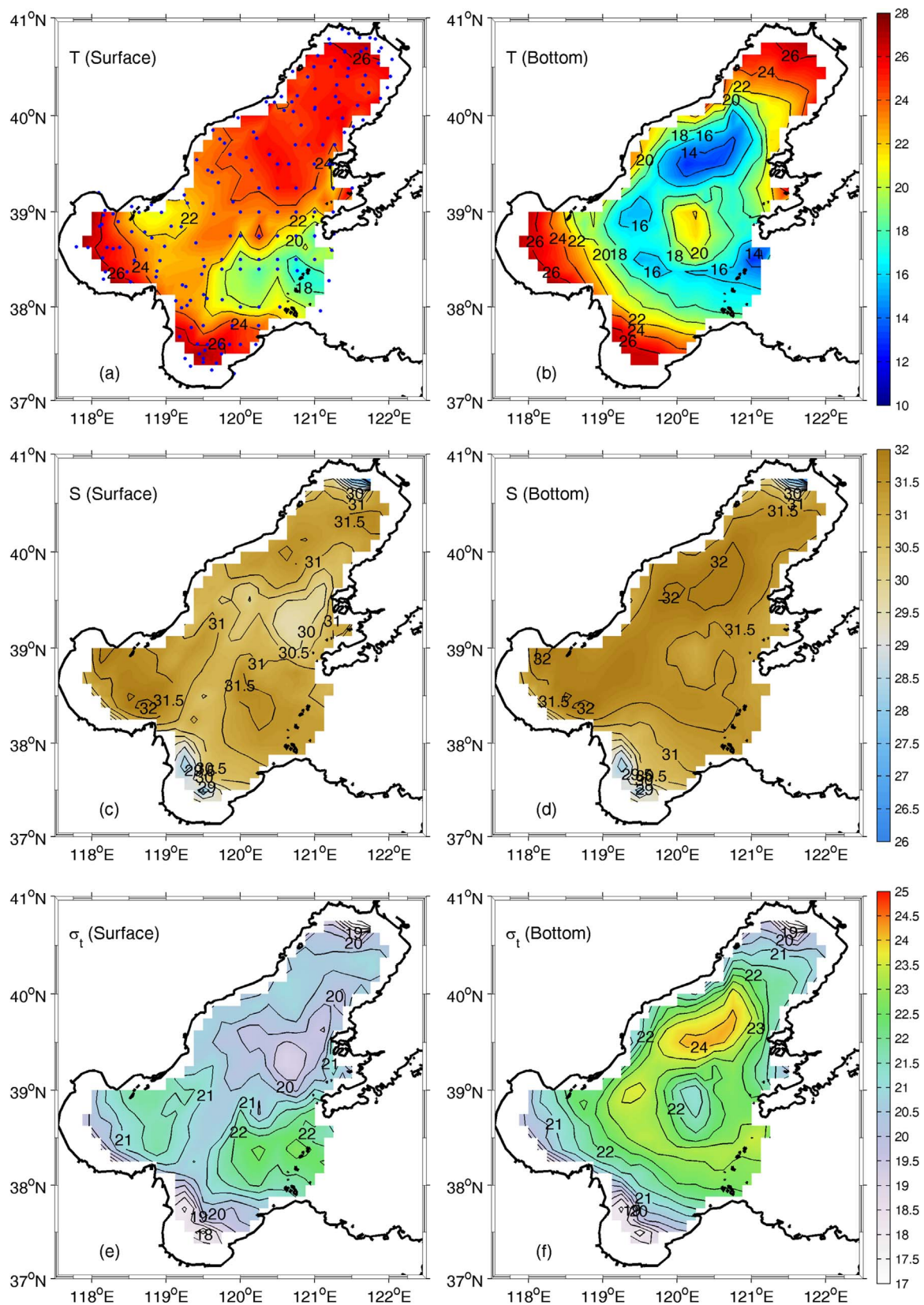


Fig. 3. Same as Fig. 2, except based on the intensive hydrographic surveys during July 19 to August 12, 2006.

features in both survey periods. Note that the cold pool density (σ_t) was much lower in 1959 (22.5 kg/m³) than in 2006 (23.5 kg/m³), as a result of higher freshwater discharge from rivers into the basin (see Section 2.2). In addition, a cold water patch in the southern Bohai Sea

was separated by Central Bank from the northern pool in both survey periods (Fig. 2b vs 3b). In contrast, a big difference can be seen in salinity at the surface and the bottom, caused by a dramatic change in the fresh water from the Yellow River and Liao River. In 1959, the

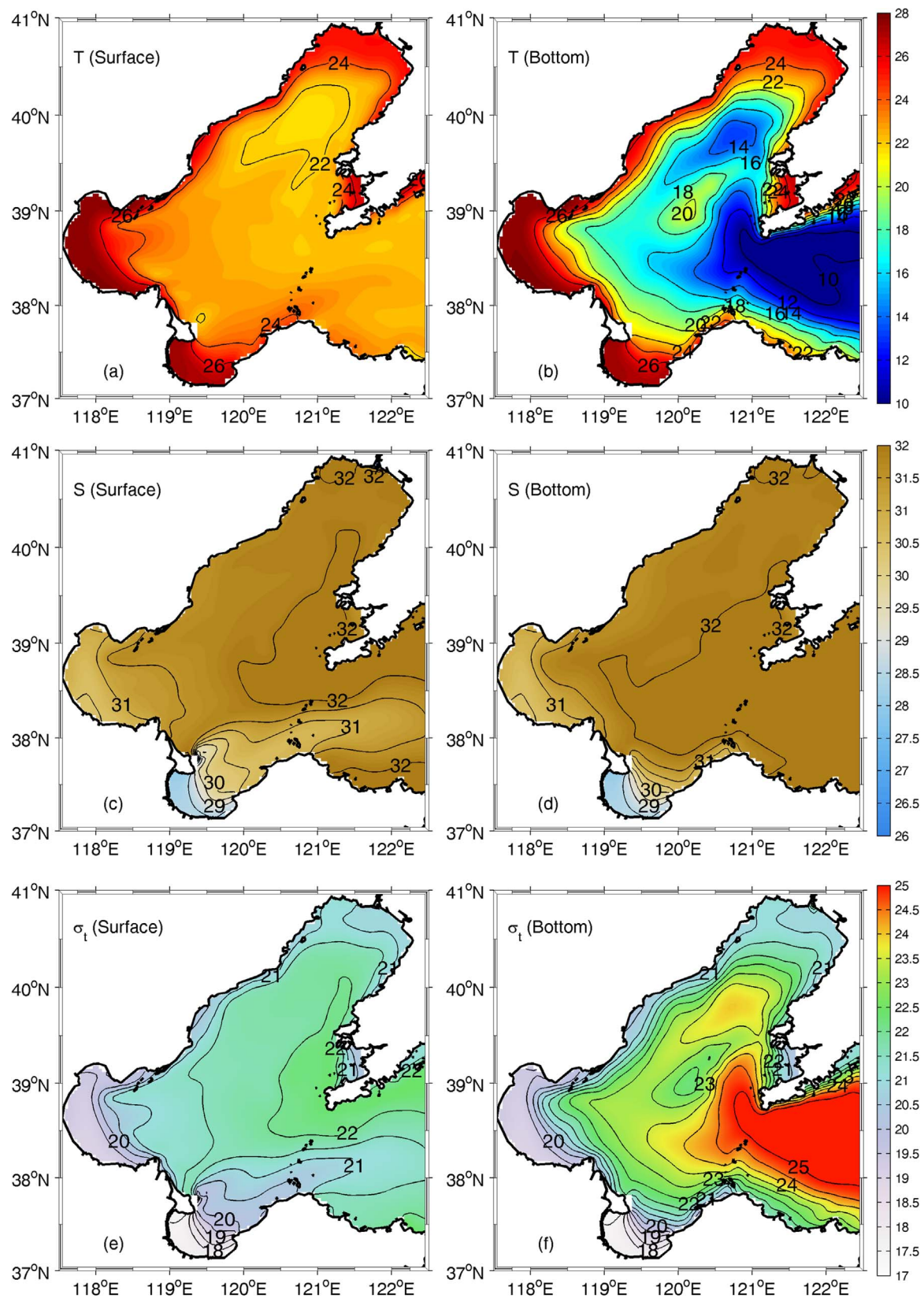


Fig. 4. Same as Fig. 3, except for modeled temperature, salinity and density in July.

outflow from the Yellow River reduced the mean salinity nearby significantly (Fig. 2c–d), while its influence was much weaker in 2006 (Fig. 3c–d). The mean salinity over the basin had increased from 30.26 to 30.47 at the surface and from 30.75 to 30.84 at the bottom.

The huge amount of sediment deposit carried by the Yellow River had also caused its delta to change shape numerous times (Yu, 2002); ultimately, the river mouth had shifted into Laizhou Bay along a man-made channel since 1976 (Huang and Su, 2002). Therefore, the Yellow

River plume was not only weak, but also changed its direction, spreading eastward in 2006.

3.2. Simulation of cold pools

Temperature and salinity in July from the reference run are reasonably consistent with observations in terms of spatial patterns

(Fig. 4). At the surface, the model reproduces higher temperatures at the tips of the three bays, which are higher than those above the cold pools (Fig. 4a). At the bottom, similar cold and dense pools can be found in Liaodong Bay, the southern Bohai Sea and the northern Yellow Sea (Fig. 4b). At Central Bank, the simulated temperature at the bottom is warmer than that of the ambient waters (Fig. 4b). The modeled horizontal gradients of temperature and density at the bottom

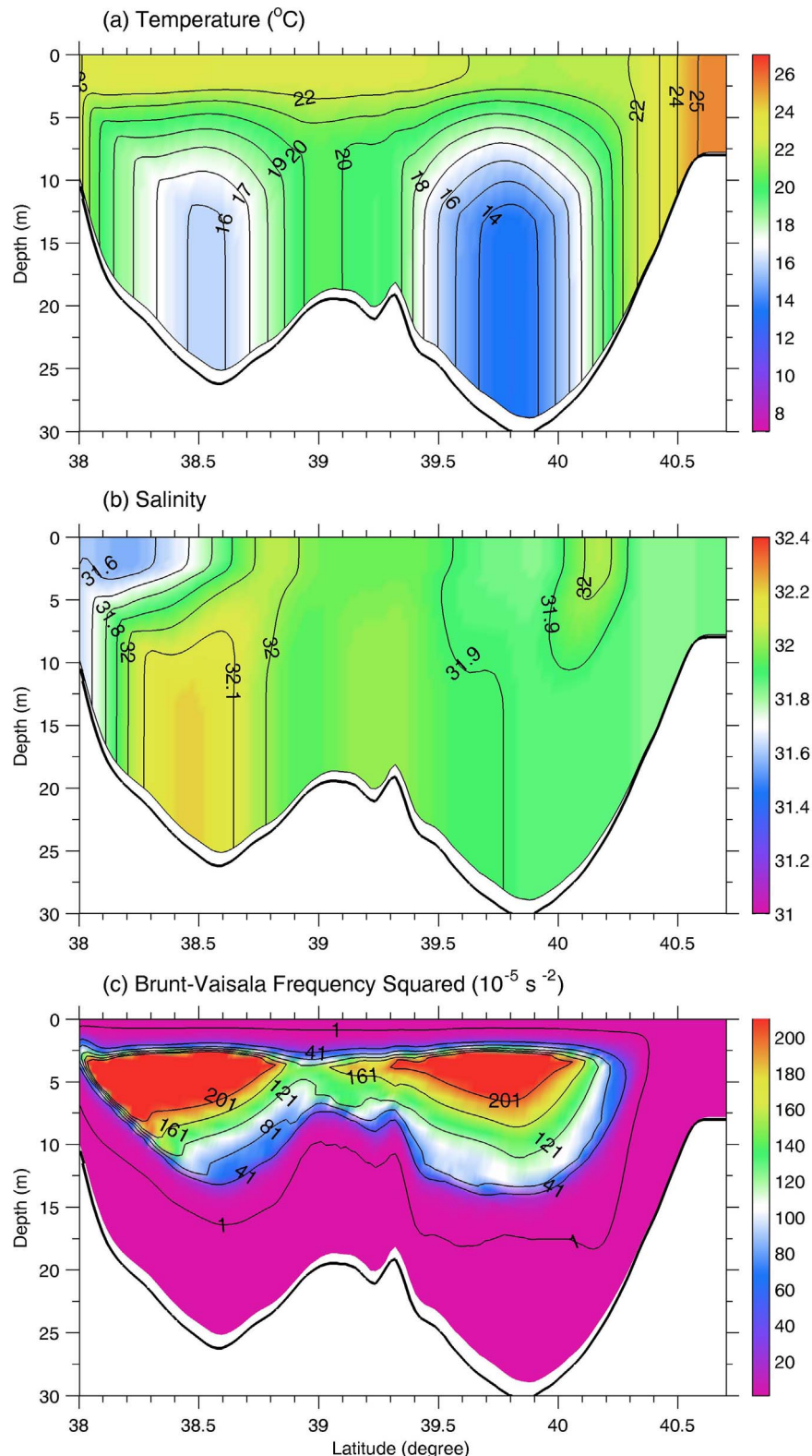


Fig. 5. Simulated monthly mean values of temperature, salinity and calculated N^2 (N : Brunt-Väisälä Frequency) along the transect (see Fig. 1) in July.

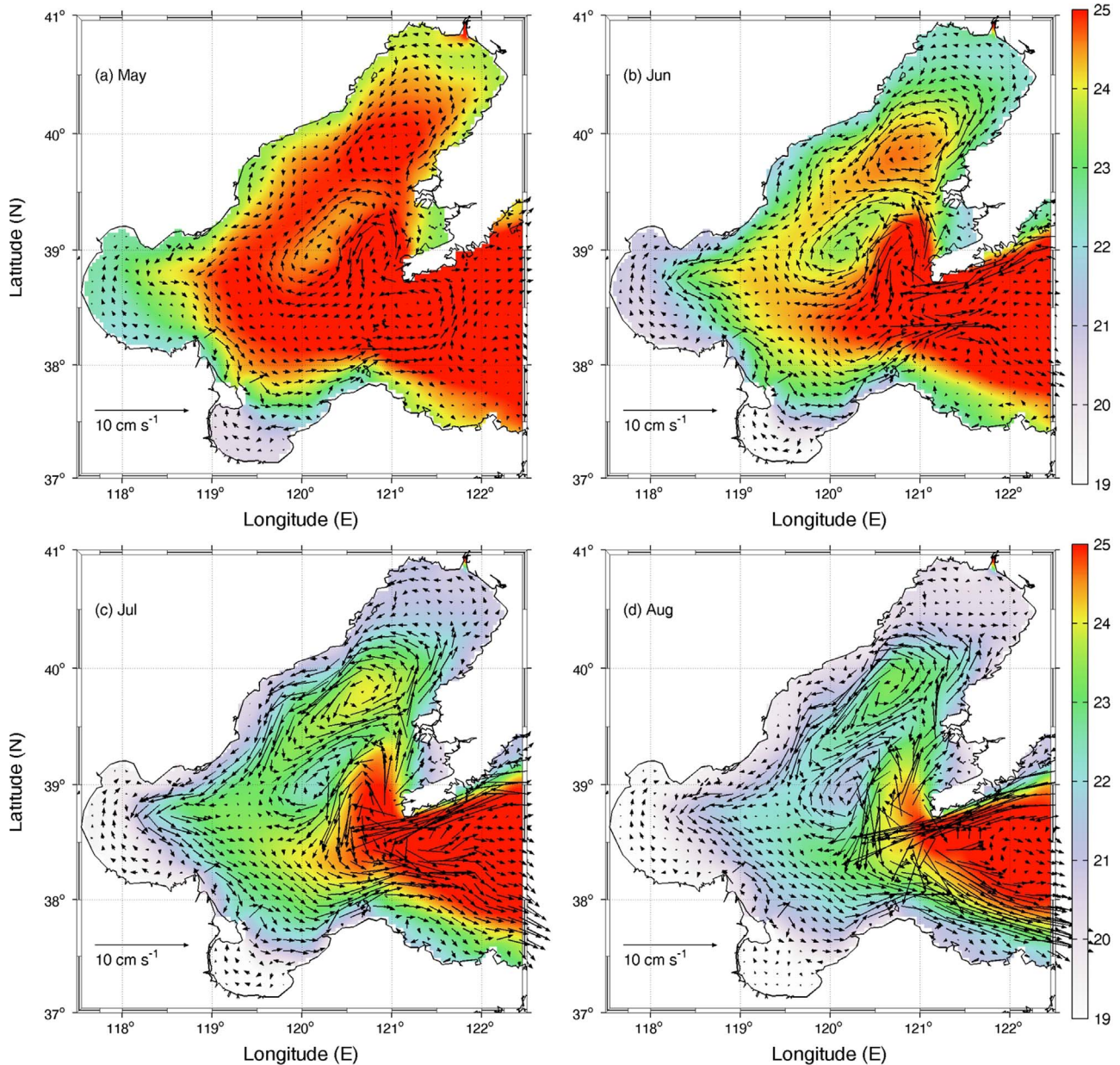


Fig. 6. Depth-averaged currents during typical stratified periods of May–August. The color scale represents the monthly-mean σ_t at the bottom. Each plotted current vector is the average of nine neighboring grids. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article).

are significant over the entire basin along the coast, which resemble those of the observations. The most significant lower salinity occurs in Laizhou Bay, followed by that in Bohai Bay (Fig. 4c–d). Salinity is higher in the eastern portion of the central Bohai Sea, probably associated with saline water intrusion from the northern Yellow Sea. The salinity difference between the surface and bottom is relatively weak compared to the temperature difference. The density pattern at the surface is more similar to that of salinity, while its pattern at the bottom looks more like that of temperature (Fig. 4e–f).

Fig. 5 shows the modelled temperature and salinity along the transect (see Fig. 1) in July based on the reference run. The transect was as an indicator of the cold pools in previous studies, e.g., Liu et al. (2003) and Lin et al. (2006). The model reproduces the two-dome structure of temperature above the depression. The stability expressed by the squared Brunt–Väisälä Frequency indicates the static feature of water column above the cold pools.

3.3. Simulation of circulation

The typical basin-scale gyre during the stratified season is represented by depth-averaged currents from May to August in the model (Fig. 6). A sub-basin-scale cyclonic gyre is developed in the northern Bohai Sea, along with an anticyclonic gyre in Central Bank during these months, both of which are aligned with the 25-m isobath. The evolution of depth-averaged flow pattern is coincident with the corresponding density field. The flow around the cyclonic gyre increases as the lateral density gradient increases. The cyclonic basin-scale gyre, primarily along the 20-m isobath, appears in May and becomes more evident after June.

The bottom density distribution shows that dense water has already formed in May, mostly around the 25-m depression. In other words, the water in the depression remains colder than its ambient water, leading to a dense pool at depth. Additionally, the cold pool is also replenished by intruding cold water from the northern Yellow Sea. At the bottom, the two tongues of dense water suggest that inflow water is

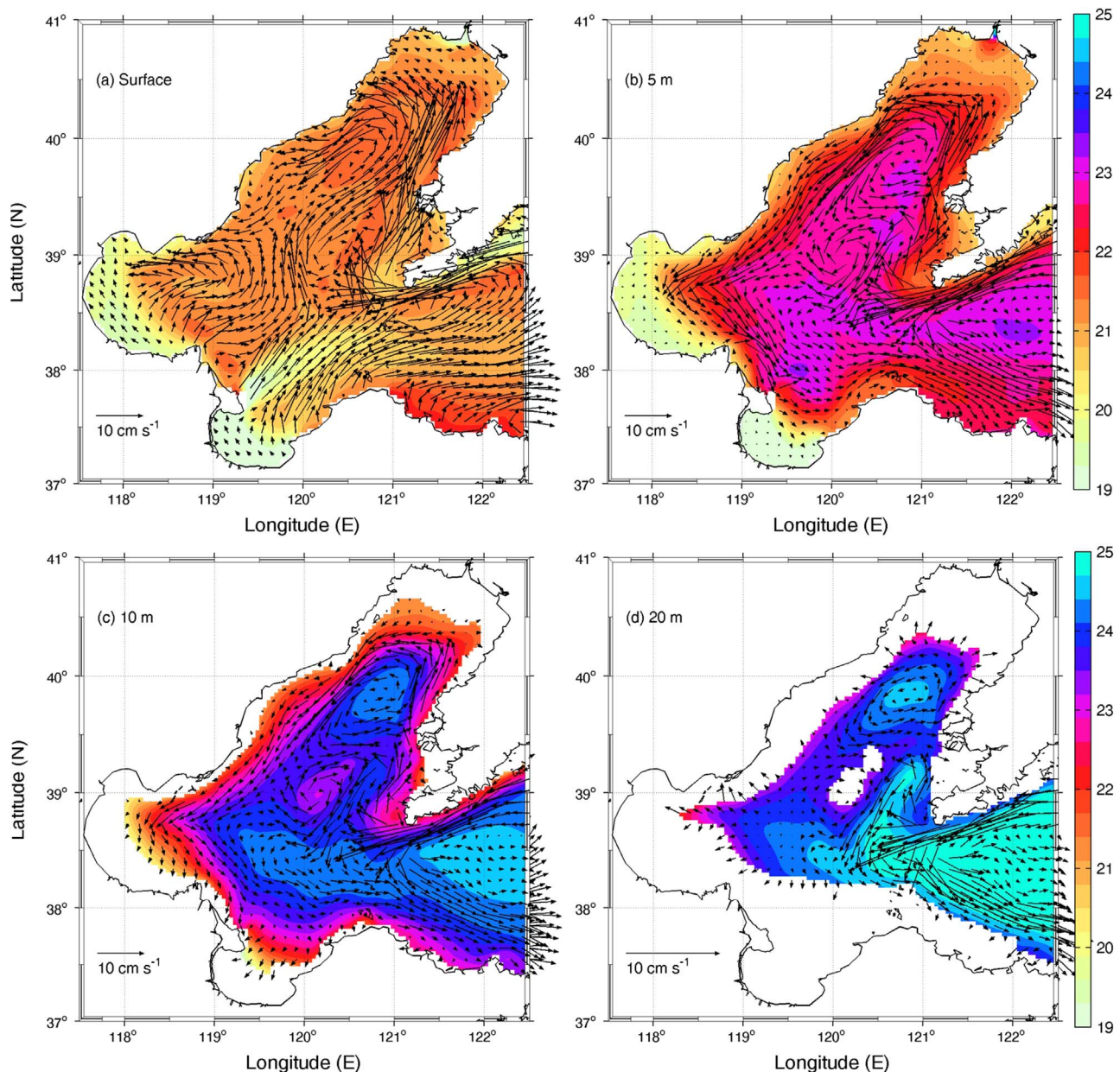


Fig. 7. Vertical structure of circulations averaged within summer (June–August) based on the reference run. (a) Surface, (b) 5 m, (c) 10 m, and (d) 20 m. Tides, wind, rivers, and baroclinic effects are all included in this run. Each plotted current vector is averaged from nine neighboring grids and shaded. Note that the scale used for plotting the flows at 20 m is twice of the scale used for the others.

separated by Central Bank with one branch entering the northern Bohai Sea and the other flowing westward.

Summer-mean circulations in different layers are used to illustrate strong baroclinic features of the basin-scale gyre (Fig. 7). A few eddy-like structures form at the surface, in addition to an eminent north-eastward flow. One cyclone is in Liaodong Bay, which is demonstrated by a strong northeastward flow east of the bay along with a relatively weaker southwestward flow west of the bay. The other cyclone of 50-km radius is located outside Bohai Bay. The mean surface current is 5.0 cm/s in the Bohai Sea, while that in the regions deeper than 20 m averages to 6.4 cm/s.

At 5 and 10 m, the circulations are notably different from those at the surface (Fig. 7b–c). A basin-scale alongshore current appears and a few eddies form in the northern basin. Eddy structure is highlighted by a cyclone in Liaodong Bay, similar to that at the surface, and a companion weak anticyclone above the central region. Another cyclone occurs northwest of the strait. Moreover, the near-bottom current

pattern is characterized by a cyclone in Liaodong Bay and another northwest of the strait (Fig. 7d).

In Figs. 6 and 7, the location of eddy-like flow coincides with the dense pool, which agrees with the observations shown in Figs. 2 and 3 despite that they apparently have different river runoffs. The flow structure is probably associated with lateral temperature gradient rather than lateral salinity gradient.

3.4. Relative contributions of different forcing factors

Five experiments are used to compare with the reference run. Each illustrates the contribution of different forcing, namely, the exchange across the strait, wind, heat flux, tide, and rivers.

Fig. 8 shows the circulations at the surface and 20 m in July, driven only by the exchange across the Bohai Strait (Case OBC). The circulation at the surface is characterized by a basin-scale cyclonic flow with a few sub-basin-scale, eddy-like structures (Fig. 8a). The

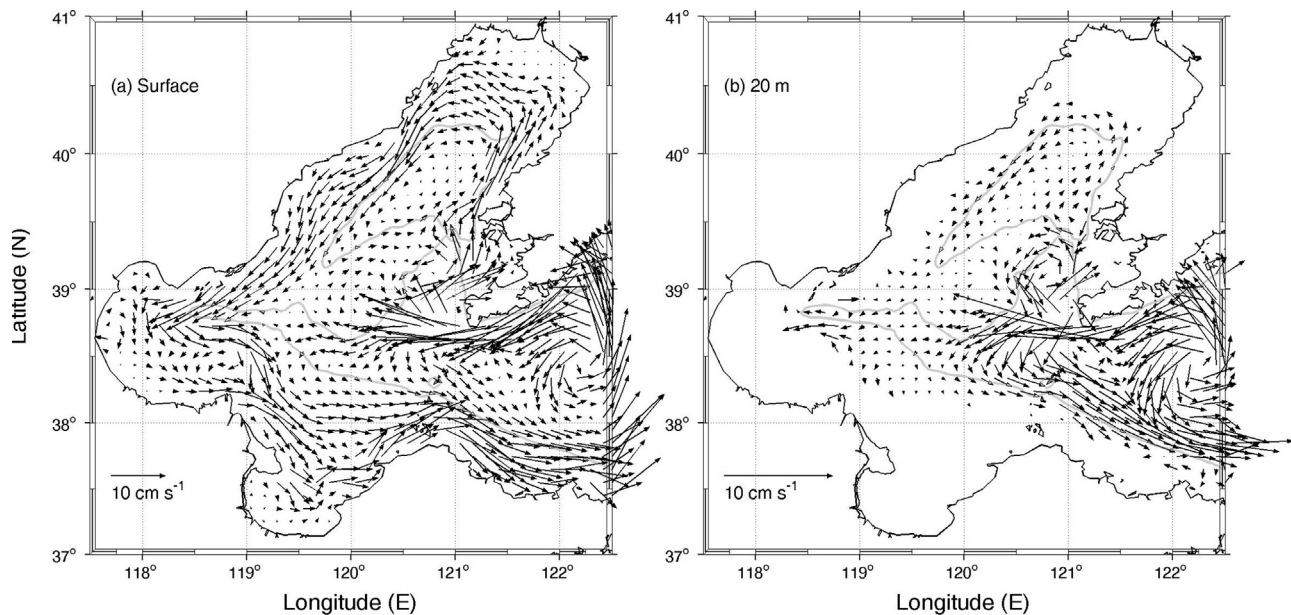


Fig. 8. Vertical structure of circulations averaged within summer (June–August) based on the sensitivity experiment of Case OBC. (a) Surface, and (b) 20 m. Each plotted current vector is averaged from nine neighboring grids. Note that the scale used for plotting the flows at 20 m is twice of the scale used for the surface.

inflow north of the strait and outflow south of the strait indicate dynamic exchanges between the Bohai Sea and the Yellow Sea. The main portion of the intruding Yellow Sea water enters Liaodong Bay along Laotieshan Channel and forms a prominent cyclonic gyre, which distributes mainly in nearshore regions shallower than 25 m. The other portion moves westward south of Central Bank, but turns eastward and joins the cyclonic gyre at places west of the strait or west of Bohai Bay. A weak cyclonic flow in Liaodong Bay and a weak anticyclonic flow are discernable despite the neglect of heat flux, indicating that the exchange across the strait contributes to the cold pool in the Bohai Sea by carrying in cold water from the northern Yellow Sea. Note that the inflow through the strait is narrow, while the outflow is relatively wider at the surface and 20-m depth (Fig. 8b).

The surface flow pattern in Case W+D+R is similar to that in the reference run, but the current speeds east and north of Liaodong Bay decrease significantly with respect to the reference run (Fig. 9a). Both the inflow north of the Bohai Strait and the outflow south of the strait are much stronger when M_2 is considered (Fig. 9b). The discernable difference in the southern Bohai Sea can also be seen at the entrance to Bohai and Laizhou bays at the edges of stratified regions, where the water depth is shallow (15 m). In contrast, no major change is observed south of QHD between 39 and 40°N and southwest of Central Bank; this is attributed to the amphidromic point for M_2 outside QHD (Fang and Yang, 1985). At 5 m, the cyclonic flow in Liaodong Bay becomes less visible compared to that in the reference run, illustrating the significance of tidal effects on the cold pools and associated circulations (not shown).

Without wind and river forcing, the surface flow field in Case T+D shows a basin-scale cyclonic gyre mostly along the 20-m isobath, cyclonic flows in Liaodong Bay and anticyclonic flows in Central Bank (Fig. 9c), resembling those at 5 m in the reference run (Fig. 7b). Fig. 9d shows the differences between Case T+D and Case T+W+D at the surface, indicating that wind effects mostly contribute to the basin-scale northeast flow at depths shallower than 5 m. Furthermore, the weak northwestward flow can be found in the three bays (Fig. 9d), but is absent from Case T+D (Fig. 9c) and tidal effects (Fig. 9b).

The difference between Case T+W+D and the reference run is hard to detect, except for the places near the river mouths, indicating that river effects are limited (Fig. 9e–f). The diluted water of the Yellow River introduces a weak northeastward flow outside Laizhou Bay, and is more significant than that of the Liao River. This less-saline water

mostly flows eastward before entering the Yellow Sea at the middle part of the strait around 38°N, and induces a weak inflow to its south. In addition, a rather weak cyclonic flow exists around Central Bank, which is related to river effects. Comparatively, the diluted water from the Liao River is mostly transported to the west.

Based on an analysis of summertime circulations and their relationship with different forcing, Fig. 10 presents schematic circulations in the Bohai Sea. The main component of the gyre includes a basin-scale cyclonic baroclinic circulation, mainly driven by the exchange across the strait (Fig. 8a) and by heat flux (Fig. 9c). There are also a few eddy-like flow structures, in Liaodong Bay, around Central Bank and at the Bohai Strait. The cyclonic circulation around the cold pool in Liaodong Bay is significant, which is generated by both heat flux and tidal mixing (Fig. 9b–c).

3.5. Lagrangian particle-tracking experiments

Circulations associated with the cold pools in the Bohai Sea have significant intra-seasonal variability (Fig. 6). The LPT experiments are aimed to compare the number of particles released during different periods, i.e., February 1 (winter) when the cold pools are absent and July 1 (summer) when stratification is significant.

Fig. 11 shows the time evolution of percentages of particles released in the Bohai Sea. These particles were released from three different layers, which differ in summer but are similar in winter in terms of retention rate. In summer, the surface layer is critical for trapping particles. Close to 95% of the particles released at surface remain in the Bohai Sea after 60 days. The loss of particles released at 10 m and that near the sea bed are similar, being 10–20% more than that at the surface. Fig. 11a also implies that the circulations in summer have a significant three-dimensional structure (see Fig. 7) by showing different capability of trapping particles at three layers. In winter, the majority of the particles remain in the Bohai Sea after 60 days, with only a 2% loss of all particles (Fig. 11b). Note that the time series of particle retention in the three layers in winter resemble each other, suggesting a nearly-homogeneous vertical flow structure in comparison with that of summer. In addition, a considerable number of particles appeared along the western coast and outside of Bohai Bay during summer in the presence of the shelf-sea front, which is different from the particles released in winter when the particles are more evenly distributed (Fig. 11c–d). This is because the particles released in

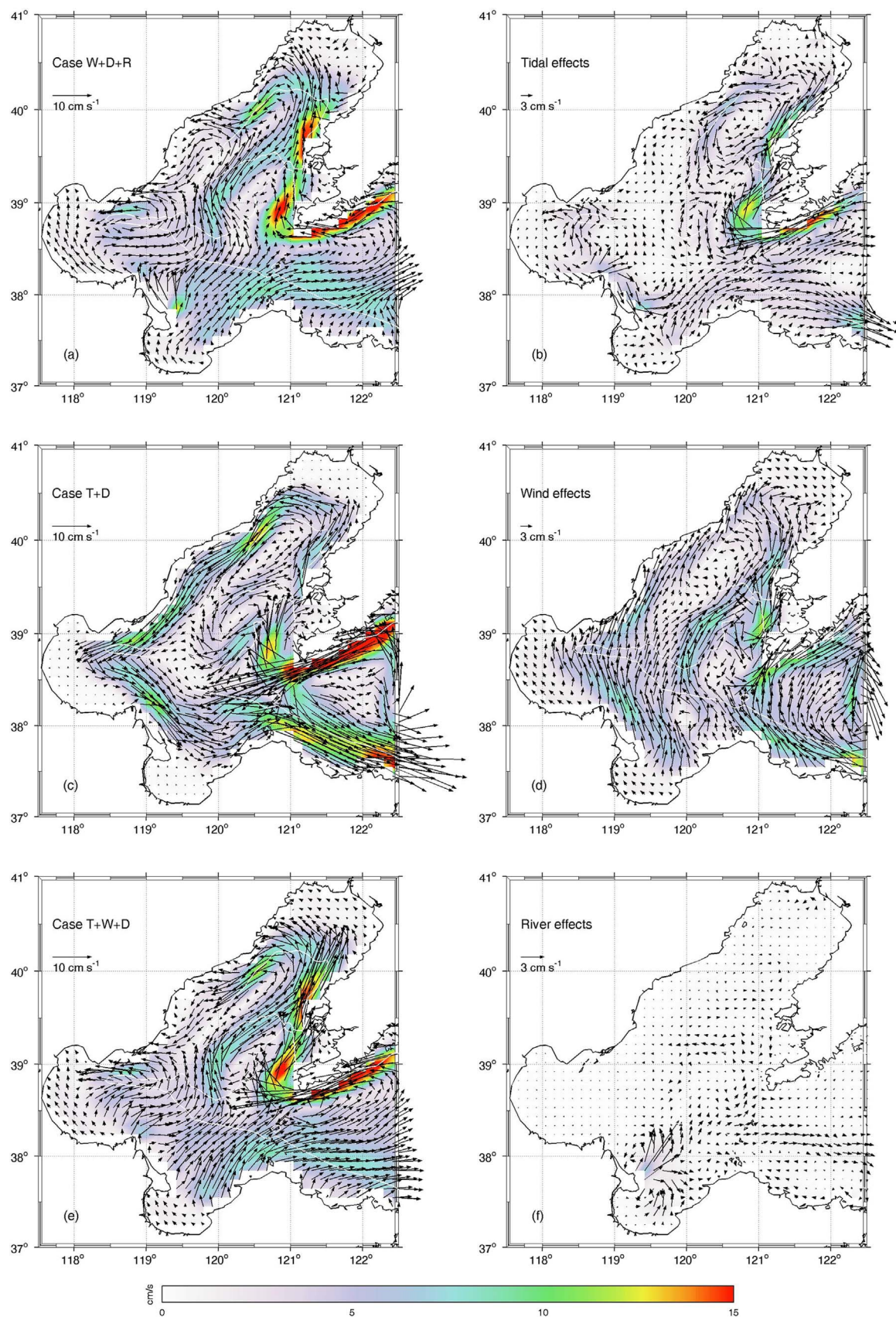


Fig. 9. Surface flows from sensitivity tests used to identify contributions of different forcing. D: Baroclinic effects; R: river runoff; T: tide; and W: wind. (a) Case W+D+R is the same as the reference run, except that M_2 tide forcing is removed. (c) Case T+D is the same as the reference run, except that both wind and rivers are removed. (e) Case T+W+D is the same as the reference run, except that the rivers around the Bohai Sea are removed.

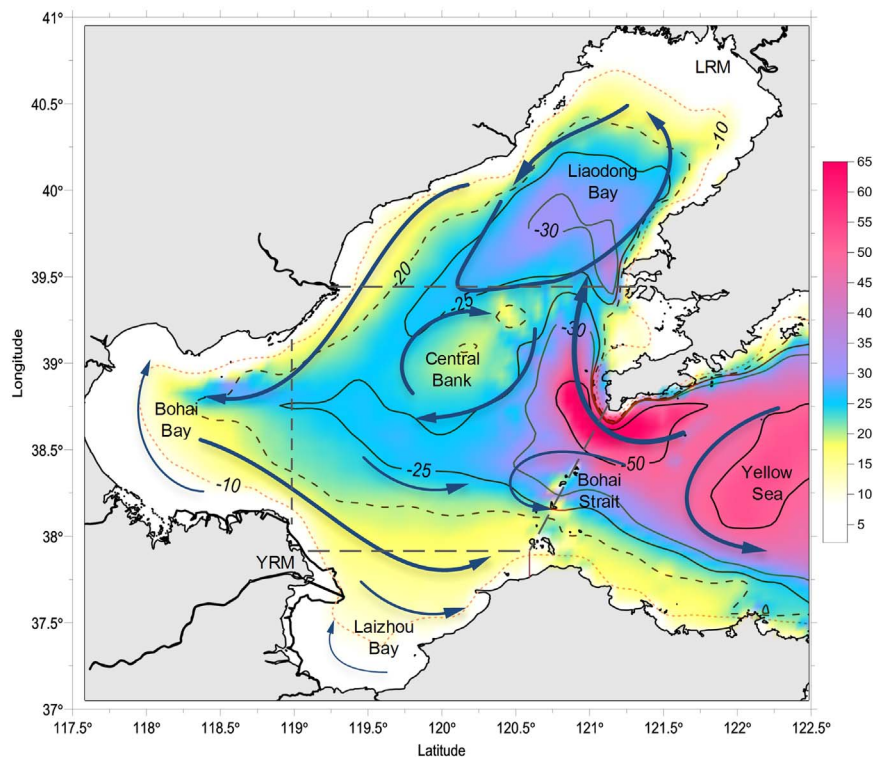


Fig. 10. Schematic of circulations in the Bohai Sea in summer based on climatological simulations. The relative magnitude of depth-averaged circulation is approximated by the width of the line. Dashed lines separate three bays from the central region, consistent with previous studies, such as Liu et al. (2012). Topographic features are shaded with superimposed isobaths.

summer are driven by cyclonic circulations and trapped in the convergence zones, both of which are associated with the fronts as suggested by field observations and the simulations of the western Irish Sea (Hill et al., 1993).

4. Discussion and conclusions

The study provides the observational evidence for the cold pools in the Bohai Sea during the summer with two large-scale surveys carried out in 1959 and 2006, respectively. The two cold pools, one in Liaodong Bay and the other in the southern Bohai Sea, were separated by Central Bank; the pattern was found rather stable during both years despite significant differences in salinity distribution.

Previous studies suggested the importance of baroclinic structure to the basin-scale gyre (Liang et al., 2006; Wang et al., 2010) and the importance of anti-cyclonic flow in the central Bohai Sea (Wang et al., 2004). In comparison, our climatological simulations reveal that the Bohai Sea circulation is characterized by a basin-scale cyclonic gyre and a few co-existing eddy-like structures from May to August, e.g., cyclonic circulations around the cold pools in Liaodong Bay and northwest of the strait, and an anticyclonic circulation above Central Bank. The circulations in Central Bank and Liaodong Bay are quite different from those in Zhao et al. (1995) (Fig. 1 vs 10). Sensitivity experiments revealed that the basic flow structure is produced mainly by the cross-strait exchange and heat flux, followed by the contribution of tide mixing. LPT experiments confirm convergence at the surface in the western Bohai Sea, which result in an intense concentration of particles released in summer. In comparison, particle retention in the three layers resembles each other in winter when water is vertically well mixed.

The modeled pattern of temperature and salinity, hence the baroclinic currents, showed some disagreement with observations. Further studies should apply highly time-variable air-sea fluxes to include realistic variability into the forcing. The cold pools have high

temporal and spatial variability because of the effects of tidal mixing, wind mixing, and the cross-strait exchanges. It is expected that synoptic baroclinic circulations be differ substantially from those forced by the monthly-mean forcing. Varying wind forcing may also change the intensity of the exchange across the strait, particular at the surface.

Both the temporal resolution and spatial resolution of the model could be improved; nevertheless the broad features of the baroclinic gyre circulation are replicated. The horizontal resolution is approximately 4 km, which is slightly larger than the first baroclinic Rossby radius of deformation (~2–3 km) in the Bohai Sea. This means that mesoscale and sub-mesoscale activities are not resolved by the current model. The mismatch of stratification between simulations and observations could have resulted from the heat flux used in this study, which is also low in spatial resolutions. Though the 30 layers in the vertical decreases the likelihood of interpolation error in the LPT experiment, the performance of random walk could be improved with a higher resolution.

In general, cold pools are likely persistent features in the Bohai Sea in summer, based on the two surveys in 1959 and 2006, along with many other field measurements. The discovery of cyclonic circulations around cold pools has important environmental implications, for the retention system affects the transports of contaminants and waste disposals (Hill et al., 1997). Similar threats to the semi-enclosed Bohai Sea become more severe in recent decades with increasing industrial and human activities around the region. The retention ability of the quasi-static cold pools is likely responsible for the low oxygen level west of Central Bank (Zhai et al., 2012), where could lead to hypoxia.

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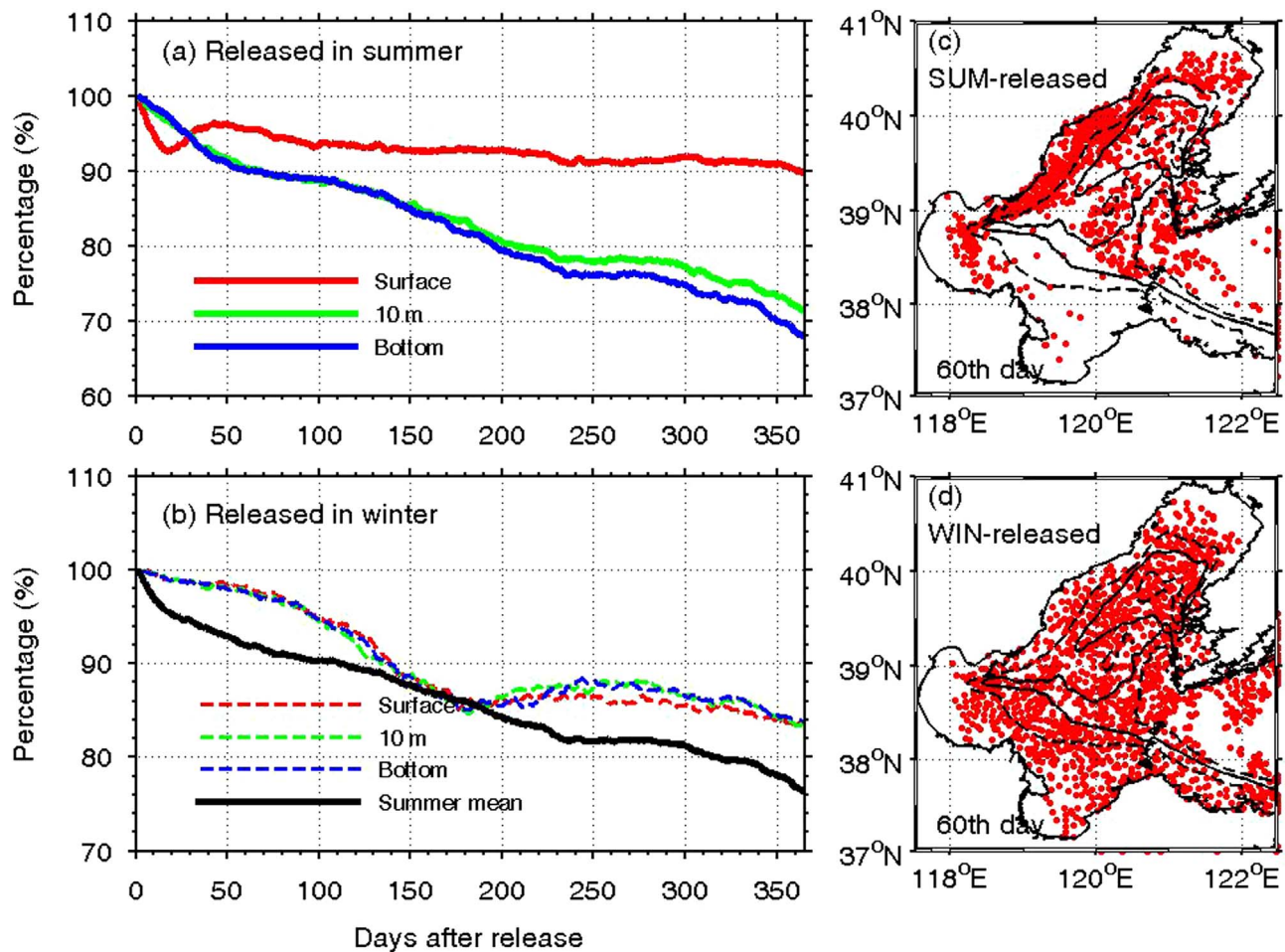


Fig. 11. Percentage of Lagrangian particles retained in the Bohai Sea (a and b) and distributions of Lagrangian particles released at the surface at the 60th day (c and d). These particles were evenly distributed initially at the surface, 10-m depth and near bed, and were released on July 1 (summer; a and c) and February 1 (winter; b and d). The percentage is the ratio of the number of particles with respect to the initial number.

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